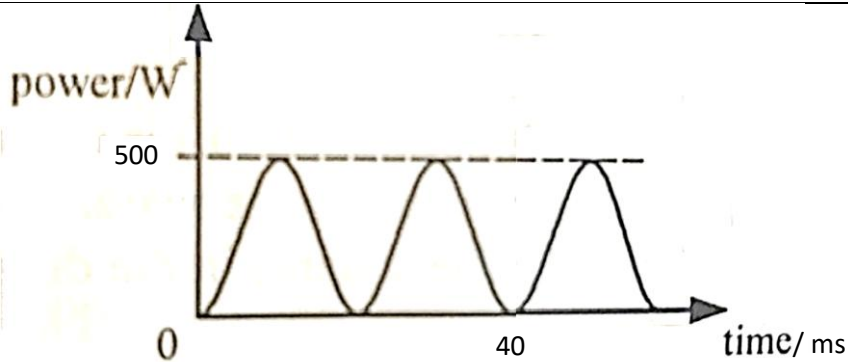


5(a)(i)	The root-mean-square value of <u>an alternating current is the equivalent value of a steady direct current that will dissipate heat at the same average rate</u> as the alternating current, in a given resistor.	B1 B1
5(a)(ii)	Frequency = $1 / (40 \times 10^{-3})$ = 25 Hz	A1
5(a)(iii)	Peak value = 10 A	A1
5(a)(iv)	Root-mean-square value of current = $10 / \sqrt{2}$ = 7.07 A	A1
5(a)(v)	 Both axes are labelled sufficiently Shape of graph is correct	A1 A1
5(b)(i)	An ideal transformer is a device that steps up or steps down the voltage of an alternating supply without power loss.	B1
5(b)(ii)	$N_p/N_s = I_s/I_p$ $300/6000 = I_s/10$ $I_s = 0.5 \text{ A}$	M1 A1